



ELF: Educational LLM Framework of Improving and Evaluating AI Generated Content for Classroom Teaching

KEHUI TAN, South China Normal University, Guangzhou, China

JIAYANG YAO, South China Normal University, Guangzhou, China

TIANQI PANG, South China Normal University, Guangzhou, China

CHENYOU FAN*, South China Normal University, Guangzhou, China

YU SONG*, South China Normal University, Guangzhou, China

Recent studies [48, 72] have demonstrated that Large Language Models (LLMs), like ChatGPT [3, 46] and LLAMA [59], can assist with routine teaching tasks and have the potential to revolutionize traditional education. However, other studies [35] highlight that LLMs often contain inaccuracies and demonstrate limited effectiveness in educational contexts. To address this issue, we propose a unified Education LLM Framework (*ELF*) that integrates LLM into classroom teaching practice to enrich high-quality dialogical content and teacher-student interactions. Unlike complex data-driven models that require vast amounts of data, our framework can quickly enhance educational engagement and teaching strategies by utilizing a few carefully selected teaching examples from master teachers with our prompting techniques. We focus on two typical classroom teaching scenarios that require AI-generated content: Dialogue Completion and Expertise Transfer Learning. The former scenario requires generating contextually appropriate dialogues, while the latter scenario requires migrating the instructional styles and organization to new teaching topics. We demonstrate the effectiveness of our data quality-centered approach in generating semantically clear and factually accurate content as organized instructions for teaching materials. We comprehensively evaluate these materials by utilizing Perplexity-based Statistical Evaluation, Human Evaluation with Questionnaires (HEQ), BertScore, Rouge, and BLEU. Experiments on two self-collected datasets show that our method significantly improves various metrics in Dialogue Completion and Expertise Transfer Learning tasks, enhancing the overall utility of AI for educational purposes.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; **Discourse, dialogue and pragmatics**; *Information extraction*; • **Applied computing** → *E-learning*; **Computer-assisted instruction**; *Interactive learning environments*.

Additional Key Words and Phrases: Envision Artificial Intelligence (AI) for Education, AI in Classroom Teaching, AI Generated Content (AIGC), Dialogue Generation, Human-in-the-loop AI, Generation Evaluation.

*Chenyou Fan and Yu Song Co-correspond.

Authors' Contact Information: Kehui Tan, South China Normal University, Guangzhou, China; e-mail: kehui.tt@gmail.com; Jiayang Yao, South China Normal University, Guangzhou, China; e-mail: 2285837953@qq.com; tianqi pang, South China Normal University, Guangzhou, China; e-mail: 2022024954@m.scnu.edu.cn; Chenyou Fan, South China Normal University, Guangzhou, China; e-mail: fanchenyou@gmail.com; Yu Song, South China Normal University, Guangzhou, China; e-mail: sungyuepu@foxmail.com.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 1936-1963/2025/2-ART

<https://doi.org/10.1145/3712065>

1 Introduction

Classroom teaching is a crucial social process that involves extended interactions between teachers and students. Within this context, teacher-student dialogues promote the exchange of knowledge, skills, and ideas, and demonstrate cognitive processes such as questioning, critical thinking, answering, reflecting, and receiving feedback. These dialogues serve as a vital medium for effective communication and the development of independent thinking skills among students.

Classroom dialogues, characterized by their complexity, present formidable challenges for processing and utilization for traditional machine learning methods. This complexity arises from their inherent conversational nature, the diverse constraints, individual teaching styles, and the difficulty in establishing quantifiable evaluation standards. The processing of such conversational texts demands a high degree of professional knowledge and pedagogical insight to accurately capture contextual continuity, maintain to language standards, and align with educational objectives.

In this paper, we explore recent advancements in Artificial Intelligence for education purposes and focus on utilizing the emerging Large Language Models (LLMs) for improving classroom teaching by generating meaningful and creative dialogues between teachers and students and providing benchmarks for evaluation. Our commitment lies in the development and application of advanced language model technologies, aiming to more effectively process and generalize limited classroom dialogue data from human experts to various topics and subjects.

The recent progress in large language models (LLMs), exemplified by the GPT series [8, 46], has significantly revolutionized both academic and practical approaches to natural language processing (NLP) tasks. AI-generated content, often abbreviated as AI-generated content (AIGC), has become prevalent in various NLP and Vision tasks, including the creation of conversational dialogues and artistic images [1].

Nevertheless, the application of AIGC for educational purposes remains an under-explored task. This task is particularly critical in primary and high school classrooms in remote areas where qualified teachers are scarce. While classroom teaching videos and materials are available online, the cost of educating experienced teachers is often prohibitively high. This prompts us to consider *whether AIGC can be leveraged to create classroom teaching dialogues, aiming to enhance student education and support teachers simultaneously.*

In this paper, *we explore and verify the potential of employing LLMs to enhance the quality of classroom teaching (CT) and generate AIGC to supplement teaching resources.* We focus on two teaching scenarios over two common courses and investigate the feasibility of utilizing AIGC to generate high-quality data for educational purposes.

The first scenario we explore is the automatic completion of a teaching dialogue, abbreviated as “**dialogue completion.**” This situation arises from the recognition that the dialogue between teachers and students can occasionally be disrupted due to unforeseen events. For example, technical malfunctions, such as a sudden breakdown of the projector, can impede the flow of the dialogue by depriving the teacher of presentation tools. Addressing such disruptions swiftly can be challenging for human teachers. We investigate the application of AIGC to complete missing dialogues, assisting teachers in quickly recovering from a disrupted classroom teaching scenario.

In the second scenario, we focus on the transfer of existing syllabi or teaching schemes from expert teachers to new topics, abbreviated as “**expertise transfer.**” This approach leverages the value of teaching schemes crafted by experienced educators. However, it is challenging for inexperienced teachers to quickly acquire the necessary skills and guidance for a new topic based on a specific teaching dialogue. Consequently, delivering a well-designed and effective lesson can be difficult. We pose the question of whether AIGC can be employed to transfer the teaching style from one subject to another while adhering to the original dialogue plan and maintaining the earnest dialogical styles.

In this paper, we demonstrate the superior applicability of the proposed Educational LLM framework (ELF) in the field of classroom teaching. Specifically, based on the annotated cases mentioned above, we identify two crucial and imperative areas of research: 1) Teacher-Student Dialogue Auto-Completion and 2) Cross-discipline Expertise Transfer Learning. In order to scientifically evaluate the performance of ELF in actual educational tasks, we further propose two evaluation criteria: 1) Perplexity-based Statistical Evaluation (PPL) and 2) Human Evaluation with Questionnaires (HEQ). By organically integrating LLMs and AIGCs into classroom teaching, we aim to address the opportunities and challenges brought by this innovative approach.

We will elaborate on these scenarios in Section 3. We first propose a unified framework - Educational LLM Framework (ELF) to present our proposed framework holistically. Then in Section 4, we discuss each classroom teaching task and the steps required to solve them in detail. Finally, in Section 5.3, we evaluate the quality of the content generated using two criteria.

We summarize our contributions in this paper as follows.

- (1) We consider a critical and under-explored topic in *AIGC For Education* and how to incorporate the capable large language models into classroom teaching properly and effectively.
- (2) We self-collected teaching datasets of topics including literature and mathematics from publicly available videos of expert teachers' lectures, which we would like to make public to boost future research.
- (3) We propose a unified framework *Educational LLM Framework (ELF)* which tackles the two most applicable teaching scenarios including dialogue completion and expertise transfer learning.
- (4) We propose two novel methods - Context Learning (CL) and Dynamic Role Setting (DyRS) - for offering a comprehensive solution suitable for classical teaching scenarios. CL enhances the LLM's perception and comprehension of context by better learning contextual representations, and DyRS injects identity representations into LLM to simulate dynamic roles' linguistic styles and professional knowledge.
- (5) We identify two practical methods for comprehensively assessing the quality of generated content: the Perplexity-based statistical evaluation (PPL) and the Human Evaluation with Questionnaires (HEQ).
- (6) Our experimental results verify that our enhanced AI content generation methodology for classroom teaching approaches the quality of human expert-annotated data, in certain tasks or datasets, even surpasses it. This outcome suggests a promising direction for future research.

2 Related Work

Teacher-student dialogue is a crucial medium for interaction in classroom teaching and has been extensively studied in the pedagogical field [36, 38]. Productive classroom dialogue fosters deeper understanding, critical thinking and academic achievement [28]. However, many teachers often lack the awareness and strategies necessary to effectively promote such dialogue [6]. Given these potential benefits and practical challenges, it is essential to provide templates for effective conversations. Therefore, we aim to enhance classroom teaching by generating productive dialogues through the integration of LLMs.

Dialogue assessment. Numerous pedagogical studies have sought to evaluate classroom dialogue [13], especially for mathematics and science [4, 5, 30]. Direct approaches include collecting student feedback for detailed evaluations [50, 54], or employing machine learning methods for predictive analysis [56, 57]. Researches [40] have explored various aspects, such as students' thinking processes from diverse viewpoints, dialogic practices aimed at specific outcomes [25], group interaction performance [30], and performance scoring [29].

Recent studies [58] have involved transcribing and systematically coding classroom dialogue, using lag sequential pattern mining technique to examine dialogic contributions and identify patterns in exchanges. This approach has facilitated an examination of how dialogue structure and content evolve [39, 40]. A study [55] showed that the effectiveness of coding frameworks in analyzing classroom dialogue. Given the complexity of dialogic interaction in classroom settings, study [10] introduced sequential pattern mining techniques to capture

the evolving structure of productive lessons. Additionally, research [57] explored using DNNs for the automatic classification of semantic content in classroom dialogues. *However, how to generate dialogues with automated AI tools and assess them with human-like standards is still an under-explored area.*

Large Language Models (LLMs) are advanced mathematical models that predict the statistical distribution of words in extensive corpora of human-generated text. A prominent example is ChatGPT [73], which showcases impressive conversational abilities with humans. LLMs are capable of performing various tasks based solely on natural language descriptions, eliminating the need for task-specific training [60]. Nowadays, LLMs significantly impact the AI community, particularly in NLP, where they are increasingly used as general-purpose language task solvers.

The field of LLMs has seen significant advancements in enhancing the quality of response, particularly through techniques such as *Chain-of-Thought* (CoT) [66], *Tree-of-thought* (ToT) [70] techniques and *Self-consistency* [64]. Additionally, other notable efforts have further improved LLM performance. For instance, Federated-LLM [33] introduces intermediate steps that help LLMs explore different reasoning paths before making decisions. The Self-Refine approach [37] enhances model outputs by using iterative feedback, showing that even top models can benefit from simple, test-time methods. Fine-tune-CoT [26] employs a large teacher model to generate samples that fine-tune a smaller model, often making it more effective than the teacher. Learn-Babble-Prune [7] boosts the reliability of outputs by rejecting responses without strong evidence, ensuring answers are grounded in the model's learned knowledge.

Recent surveys [9, 68, 73] have extensively discussed the advancements in improving LLMs, offering a thorough overview of various techniques and their strengths, limitations, and future potential. Rather than proposing general methods for enhancing LLM prompting or training, our focus is on improving LLMs in classroom teaching through specific techniques like context learning and dynamic role setting.

Large Language Models (LLMs) in Education. A survey [63] highlights the emergence of LLMs in various education applications, such as study assisting, teaching assisting and adaptive learning, which can be categorized based on its users' role, showing that the rapid development of LLMs has revolutionized education.

Curriculum Instruction Tuning [22] utilizes a teacher LLM to design a curriculum for student LLM training, illustrating the potential applications of LLMs in educational contexts. Abdelghani et al. [2] explore LLMs as pedagogical agents to foster curiosity-driven learning, indicating that LLMs can significantly enhance the expression of curiosity among learners. Dijkstra [15] further demonstrates the utility of LLMs in creating interactive educational materials to improve student learning outcomes and engagement. From the educators' perspective, Kasneci et al. [31] investigate LLMs's role in automating student answers assessment, providing adaptive feedback and generating instructional content. Moore et al. [43] further suggest that when LLMs are fine-tuned to specific domains, they can be powerful tools for teachers in evaluating the quality and pedagogical value of student responses. Additionally, recent discussions [51, 65, 71] have focused on the automatic generation of mathematical word problems, highlighting the potential of LLMs to understand equations and contextualize them appropriately. Collectively, these studies underscore the promise of leveraging LLMs to generate long-term content for classroom education, thereby enhancing the teaching and learning experience.

AI-Generated Content (AIGC) in Education. In recent years, the development of AI-Generated Content (AIGC) has attracted significant attention, especially with the emergence of GPT series [45, 46], DALL-E-2 [20] and DreamFusion [49]. Large-scale models become increasingly important in AIGC and can provide better intent extraction and improved generation results. Moreover, with the continuous iteration of models and methods, the distribution that models can learn becomes more comprehensive and closer to reality, thus generating more realistic and higher-quality content [9, 16].

Research [67] indicates that AIGC significantly enhances personalized and dynamic learning experiences, thereby improving teaching quality and efficiency. AIGC technology fosters content innovation in education by integrating real-time data, offering new perspectives, tools and creating a more interactive and personalized

learning environment. Research [69] demonstrates that AIGC can increase teaching efficiency to 96% and has the potential to enable personalized learning, provide customized resources and guidance, and boost students' outcomes and motivation in higher education.

AI-Generated content evaluation techniques. Researchers [11] have validated the reliability of perplexity (PPL) as a performance metric using Correlation Coefficients (ICC) [53]. PPL is considered highly reliable when the ICC value exceeds 0.9 [32]. Experiments further indicated that both N-gram-based models and GPT-2-based models achieve high ICC values (>0.8), confirming PPL's reliability. Additionally, Cui et al. [12] found that PPL is frequently used in generation-based models, highlighting its broader applicability across various datasets and tasks.

Research [12] suggests potential risks in training dialog models using evaluation metrics from other NLP tasks, as these metrics may not adequately assess dialog quality. This underscores the need to develop automatic evaluation metrics that closely align with human assessments. Tools such as WordNet [42] and paragraph vectors model [24] leverage semantic relationships and contextual information to enhance text coherence and consistency. Elkins et al. [19] introduce a framework for evaluating the quality of generated educational questions encompassing five metrics: relevance, adherence, grammar, answerability, and usefulness. The first four indicators serve as objective measures commonly utilized in question generation literature to assess issues like hallucinations or incorrect outputs [44], while the usefulness metric, being more nuanced, is assessed subjectively on a scale.

3 Framework Overview

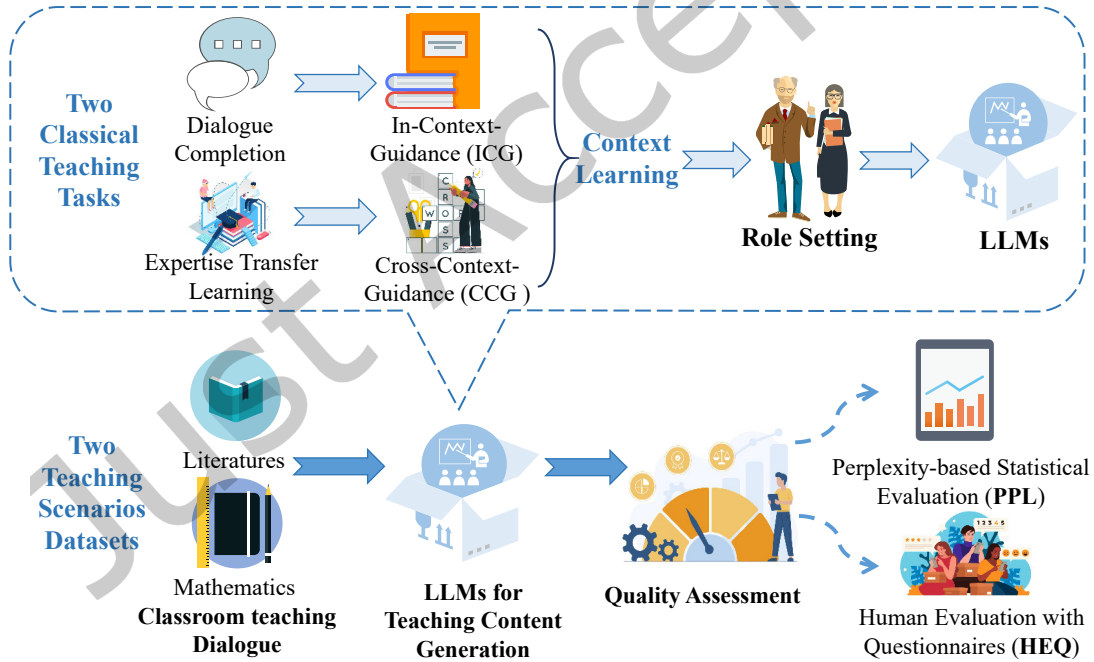


Fig. 1. Illustration of our integrated framework for implementing AI-generated content (AIGC) in classroom teaching. We investigate the improvement of content generation quality through the utilization of Large Language Models (LLMs) for Teaching Content Generation in two scenarios (Dialogue Completion and Expertise Transfer Learning). Our study encompasses two primary teaching subjects (Literature and Mathematics). We then employ two assessment methods (PPL, HEQ) to evaluate content quality.

In this section, we delve into the challenges associated with effectively generating instructional conversations using existing Large Language Models (LLMs). To address these challenges, we introduce a unified framework, as illustrated in Fig. 1, that harnesses LLMs for generating pedagogical content under two fundamental pedagogical scenarios: Dialog Completion and Expertise Transfer Learning. Moreover, our exploration spans two distinct subjects - literature and mathematics. We will first introduce our self-collected datasets in Section 3.1, and provide an overview of our techniques for improving and assessing the generated contents in Section 3.2 and Section 3.3.

3.1 Overview of our self-collected classroom teaching datasets

With the rapid advancement of technology and the flourishing of education, teachers typically record their lessons with the assistance of an Intelligent Recording and Broadcasting (IRB) system, which provides high-definition audio and video recordings. After obtaining their students' consent, these recordings are uploaded to the "National Education Resources Public Service Platform"¹, a publicly accessible and free resource. This platform offers authentic classroom dialogue data, featuring lectures by master teachers. Leveraging this resource, we transcribed high-quality audio collected through the IRB system into text, creating an experimental dataset. This access to genuine classroom interactions provides a robust foundation for AI in education research, supported by a primary corpus of classroom teaching interactions.

Considering the enormity of the dataset, we randomly sampled different samples of classroom dialogues to ensure the representativeness of the experiment. We selected 20 courses each for the topics of *Chinese Literature* and *Mathematics*, with each course lasting 40 minutes, as our experimental data. These courses span a broad range of student groups from grades one to nine and encompass diverse teaching topics. These two subjects were selected due to their representative nature, covering the basic courses of Arts and Sciences respectively, and their compulsory status in the education curriculum, thus providing valuable reference data. The dialogue scenes focus on whole-class teaching, with typical class sizes of around 40 students, as mandated by the Ministry of Education. Previous studies [27, 41] have shown that small-group dialogue is often unproductive due to students' lack of understanding of group discussion purposes and execution. Group work is rarely structured to facilitate rich exchanges of opinion, therefore we focus on whole-class dialogue. Table 1 is a detailed description of the two datasets, containing Sub-topics, Course Distribution, Dialogue Rounds per Class, and Chinese Characters per Class.

Table 1. Summary of the Chinese Literature and Mathematics Datasets

Dataset	Sub-topics	Course Distribution	Dialogue Rounds per Class	Chinese Characters per Class
Literature	Ancient poetry, prose, novels, historical stories, literary appreciation, etc.	4 primary school courses, 16 junior high school courses.	Max: 249, Min: 55, Avg: 145.55	Max: 8,264.0, Min: 4,175.0, Avg: 5,921.9
Mathematics	Algebra, geometry, probability, statistics, etc.	3 primary school courses, 17 junior high school courses.	Max: 417, Min: 124, Avg: 246.3	Max: 5,837.0, Min: 2,088.0, Avg: 4,405.9

¹<https://basic.smartedu.cn/>

Faced with such a complex and diverse array of teaching subjects, the traditional manual approach to writing lesson plans is both inefficient and difficult to ensure quality. Therefore, the introduction of AI-generated lesson plans technology has become an urgent need to solve this problem.

These datasets have played a crucial role in our research. We employ them to validate the applicability and effectiveness of our proposed methods in various subject domains. By leveraging transcriptions from instructional videos of renowned teacher-led courses, we aim to provide valuable resources and references for future educational research and technological advancements.

Table. 2 presents sample instances from our two dialogue datasets. The original dataset is in Chinese, but for readability, we have translated it into English for presentation.

Table 2. Here are two classroom dialogue dataset samples.

Example 1 (Literature)	Example 2 (Mathematics)
T: Hello, everyone. Today, we will continue our study of the renowned Tang Dynasty poet Wang Wei and ... T: First, let us contemplate the compositional context of ... S: Indeed, I agree with the teacher's views. The background of this poem ... T: Excellent, your viewpoint is highly valuable. So in this poem, what emotions does Wang Wei express? S: Wang Wei expresses the emotions of friendship... T: Indeed, you are absolutely correct. In this poem, Wang Wei conveys ... S: Teacher, I have also noticed numerous detailed descriptions in this poem, such as "The morning rain moistens ... T: Thank you, your viewpoint is valuable. These detailed descriptions ... S: Finally I want to pose a question to the teacher: What is the theme of this poem? T: The theme of this poem is separation and friendship. Through depicting ... S: I deeply appreciate the response, as this lesson has enriched my understanding. T: Thanks. You have learned a lot. In next class, we will study other poems by Wang Wei and experience the charm of his poetry.	T: Hello, may I inquire if there are any questions that you require assistance with? S: Good day, teacher. I am interested in learning about the process of integer division and how to comprehend the calculations. T: Certainly. The calculation method for integer division can be learned by starting with fractions and decimals. Firstly we can convert ... S: Understood. Teacher, I would like to understand the method for dividing a three-digit number by a one-digit number and decide the number of digits in the quotient. T: The estimation method for dividing a three-digit number by a one-digit number involves breaking down the three-digit number into two two-digit numbers and dividing each number separately resulting in quotients and remainders. ... S: You're welcome, teacher. I have another question regarding how to draw grids to better understand long division. T: Certainly. We can use graph paper to draw long division calculations. Firstly, we need to divide graph paper into several squares, with each square denoting a calculation step. Then, we can follow a specific order to perform ... S: I understand. Thanks for your explanation.

3.2 LLMs for teaching content generation

The inability of existing LLMs to target the generation of more utilizable features in terms of classroom teaching, which tend more toward correct but uninformative content, has prevented existing LLMs from playing a more creative role in this area than previously. Therefore, to enhance the quality of content generated by LLMs, we propose a unified framework that integrates LLMs, as shown at the top of Figure 1, and introduce two modules: Context Learning and Dynamic Role Setting to handle traditional teaching tasks such as dialogue completion and expert style transfer. The specific implementation will be discussed in detail in Section 4.

When dealing with vast and complex text data, especially in dialogue scenarios, existing LLMs often encounter challenges related to memory decay, which impede their ability to effectively capture global information in long texts. Moreover, as LLMs generate text based on a wide range of potential responses, variability in the style of the generated content often arises. These limitations can result in the omission of key information and insufficient attention to details, ultimately affecting the quality of output. To address this issue, we innovatively designed the “Context Learning” (CL) module, which includes two methods: In-Context-Guidance (ICG) and Cross-Context-Guidance (CCG). In-Context-Guidance is a methodology widely used to enhance the performance of LLMs by leveraging contextual information to guide their actions and decisions. For example, GraphRAG [18] summarizes all documents to facilitate mapping for retrieval, enabling the model to assess and utilize relevant context efficiently. Similarly, Reflexion [52] iteratively generates actions and responses, summarizing feedback into its memory to reflect and generate new actions. The REFLECT framework [34] guides the model by summarizing a robot’s current experiences and failure explanations, which helps in correcting its current plan. In our approach, ICG enhances the model’s ability to capture the overall context and key information of the text through a global perception mechanism. This reduces content forgetting, improves performance in processing long texts, and intelligently filters redundant details to ensure more accurate generated content. Cross-Context-Guidance was initially applied in the field of vision, as demonstrated by Image Style Transfer using convolutional networks [23], which maps the style of one image onto another with completely different content. In Natural Language Processing (NLP), a widely adopted cross-context learning technique is Chain of Thought (CoT) [66], which provides an output template for the LLM to follow. In our approach, CCG enhances the model’s ability to grasp the nuances of generation style by cross-learning from and aligning with given templates. This innovative design significantly improves the accuracy and stylistic consistency of LLMs when handling complex text tasks.

Without additional prompts, LLMs tend to generate general content lacking deeper professional-level thinking. Therefore, to enhance LLMs’ ability to generate high-quality content in the field of education, we propose the addition of the Dynamic Role Setting module. This role could be filled by a variety of professionals, such as English literature, mathematics, or physics teachers accordingly. This explicit setting of role player provides specific context for the generated content. The purpose of this module is to improve engagement and personalization by setting specific role characteristics and requirements when generating teaching content. This enhances the personalization and adaptability of the generated content, allowing the model to produce more targeted output based on the needs and role characteristics of the target audience. Simultaneously, this design can also tap into LLMs’ thinking on specific domains to some extent, enabling them to generate more professional and normative expressions. Such a module can make LLMs generate more thoughtful and professional content in the field of education, improving teaching quality and providing users with richer and more valuable educational resources.

3.3 Generate content quality assessment

To comprehensively and accurately evaluate the practical effectiveness of our proposed method in improving the quality of content generated by LLMs, we have adopted two complementary assessment approaches: statistical evaluation based on perplexity (PPL) and human evaluation based on questionnaires (HEQ). The combination of these two methods enables in-depth and detailed analysis of the quality of generated text from both quantitative statistical indicators and human perception perspectives.

As a standard evaluation metric in the field of natural language processing, perplexity is widely used to measure the degree of fit between a language model and test data. A lower perplexity value typically indicates stronger predictive capabilities of the model for text data, indirectly reflecting higher quality of generated content. We chose to use perplexity as one of our evaluation metrics precisely because it provides objective and quantifiable results, facilitating comparisons and analyses of differences between different methods. In this experiment, we employed a perplexity calculation method based on GPT-2 to assess the fluency and consistency of the text generated by the model. This approach allows us to obtain a specific numerical result, enabling more intuitive judgments about whether the quality of the model’s generated content has improved.

However, while statistical evaluation based on perplexity can provide quantitative metrics for text quality, it still struggles to fully capture human perception of text complexity. To address this limitation, we further proposed a human evaluation method based on questionnaires. Through designing comprehensive questionnaires, we collect evaluators’ feedback on the generated content from two key dimensions: semantic analysis and educational aspects. The semantic analysis dimension includes evaluations of the generated content’s language smoothness, clarity of expression, and contextual consistency. And the educational dimension includes the evaluation of factual accuracy, richness of instructional strategies, and rationality of

instructional organization. This artificial evaluation method can directly map human intuitive judgment and perception of the generated content, thus providing us with more rich and fine-grained quality evaluation information. By combining these two evaluation approaches, we can gain a more comprehensive and in-depth insight into the content generation quality improvement of LLM under our method, demonstrating our comprehensive consideration in the evaluation process and full recognition of the value of various evaluation methods.

4 Scenarios and Approaches

In this section, we will thoroughly elaborate on our proposed unified framework that integrates LLMs to tackle common teaching tasks. This unified framework initially undergoes Context Learning (including In-Context-Guidance and Cross-Context-Guidance) to ensure a comprehensive understanding of the teaching environment and content. Subsequently, by precisely introducing Dynamic Role Settings, the framework provides LLMs with more professional and relevant content output. Based on these two core modules, LLMs can generate results that are highly aligned with teaching tasks through stepwise reasoning.

We will delve into the operational modes of two primary tasks within this unified framework: Teacher-student Dialogue Auto-Completion (Section.4.1) and Expertise Transfer Learning (Section.4.2). This discussion will demonstrate how these tasks synergistically collaborate to generate superior quality teaching materials. Our solution is designed for seamless integration into classroom instruction, not only to empower teachers in enhancing classroom interaction and dynamism but also to ultimately improve teaching quality through the utilization of AIGC.

4.1 Teacher-student Dialogue Auto-Completion

4.1.1 Task Description. In the field of education, classroom dialogue is regarded as a key link in teaching interaction. It not only promotes students' thinking and stimulates their learning interest, but also deepens students' understanding and application of knowledge. However, in the actual teaching process, due to many unforeseen factors such as equipment failures and sudden teacher absences, the continuity of classroom dialogue is often affected, thereby interfering with the overall effectiveness of teaching. To address this issue, we propose a specific solution - "Teacher-student Dialogue Auto-Completion".

The creation of this teaching scenario stems from a dual focus on continuity and quality of classroom teaching. When unexpected situations interrupt classroom dialogue, in order to ensure the smooth progress of teaching, we envision building an artificial intelligence dialogue system that can temporarily replace the role of the teacher. This system needs to have the ability to understand and respond to student questions, while also being able to lead the flow of classroom dialogue to ensure the coherence and depth of knowledge transmission. This not only can significantly improve students' learning experience, but also make the entire classroom learning environment more active and efficient. Furthermore, it can also produce supplementary teaching materials, such as additional teaching points and case studies, based on existing classrooms, enriching the curriculum and providing diverse perspectives on the subject matter. For students, they can utilize it to obtain any missing teaching content, thus filling their knowledge gaps. This can be particularly useful for revision purposes or for catching up on missed classes.

In solving the problem of this task, our goal is not only limited to filling in the blanks of dialogue, but more dedicated to optimize the entire teaching process, so as to fundamentally improve teaching quality. The necessity of this measure is reflected in several aspects: First, coherent classroom teaching can better demonstrate the overall picture of knowledge and help students build a systematic and complete knowledge system; Secondly, through high-quality guidance of classroom dialogue, students can more deeply understand knowledge points, expand thinking and improve application ability; Finally, this solution can also provide powerful teaching assistance tools to help teachers more effectively manage classrooms and guide dialogue, thus further improving teaching effectiveness.

Based on the above considerations, we hope to achieve the following specific effects through the "Teacher-student Dialogue Auto-Completion" task: 1) Effectively address the problem of classroom dialogue interruptions caused by accidental events to ensure continuity and stability of teaching; 2) Improve students' learning participation and thinking depth through the guidance and assistance of an intelligent dialogue system; 3) Provide teachers with a whole new means of teaching support to help them carry out classroom teaching activities more efficiently. Therefore, we clearly define the task as: given an incomplete dialogue with gaps, generate a complete dialogue content that matches the classroom teaching scenario.

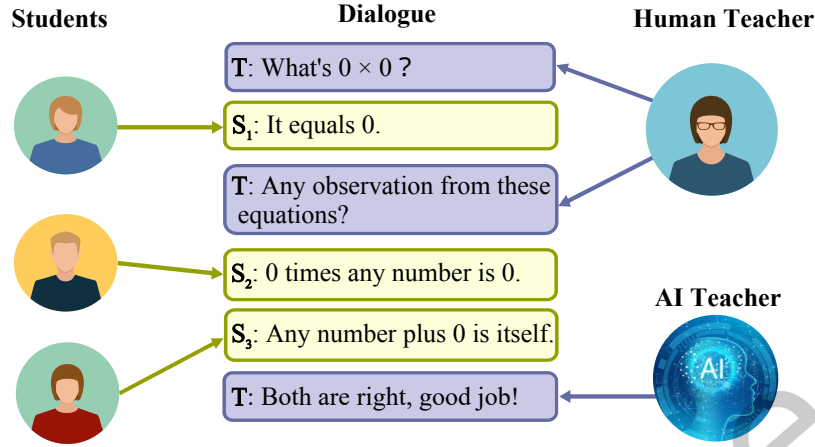


Fig. 2. An overview of classroom teaching dialogue. In our AI-aimed teaching scenarios, the teacher's conversations can be generated either by humans or AIs.

4.1.2 Experimental Setup. Our research shows that traditional zero-shot learning methods, while capable of some effectiveness, generally fail to generate dialogues of satisfactory quality when applied to the task of completing classroom dialogues. Such methods may only be able to achieve basic contiguity with the preceding context but perform poorly in terms of coherence with subsequent sections. In some cases, the model demonstrates an insufficient understanding of the task itself and directly continues the following dialogue content instead of completing the omitted portion as intended.

To address these challenges, we propose an approach called “In-Context Guidance (ICG)”. Specifically, we guide the LLM to summarize the classroom dialogue containing missing content and distill it into ideas for classroom design. This allows the model to grasp the classroom dialogue containing omissions from a holistic perspective. Experimental results show that when the LLM summarizes and distills out ideas for classroom design, it will spontaneously fill in the missing fragment content during the process of generating ideas for classroom design, indirectly self-prompting to indicate the general direction for completing the dialogue. This approach ensures not only coherence within the dialogue context but also a close match with the teacher's overall teaching design concept. To further optimize the generation of dialogue content, we introduce a role assignment strategy to fully leverage the LLM's potential.

The specific implementation process of our method is as follows:

Step 1: Input. Provide the classroom dialogue containing omitted content as input.

Step 2: Global Understanding. Use In-Context Guidance (ICG) to guide the LLM to summarize the input classroom dialogue containing gaps and distill the core ideas for classroom design. This aims to give the LLM a global understanding of the dialogue while removing redundant detail information.

Step 3: Dynamic Role Assignment. To obtain more professional and educationally valid output, we assign the LLM a specific role, such as “master teacher”. By providing the prompt “Assume you are a master teacher”, we expect the model to generate more apt and professional teaching content from this role's perspective.

Step 4: Task Explanation. Provide the complete teaching plan and identification of the incomplete fragment (such as name or number) to clearly instruct the LLM to complete the task of filling in the missing piece. This helps ensure the model has a clear understanding of the specific task to be completed.

Step 5: Generation. Based on the previous steps and guidance, we have the LLM generate the completed dialogue content.

We conducted a comparative study of this method versus traditional “zero-shot” methods. In “zero-shot” methods, we directly provide the classroom dialogue containing gaps to the LLM and prompt it to generate the missing middle portion. Through in-depth analysis of the text generated by the LLM under the two methods, we found that our proposed method significantly outperforms “zero-shot” methods in terms of accuracy, consistency of style and richness of content. The specific comparative results and in-depth analysis will be discussed in detail in Section 5.3.

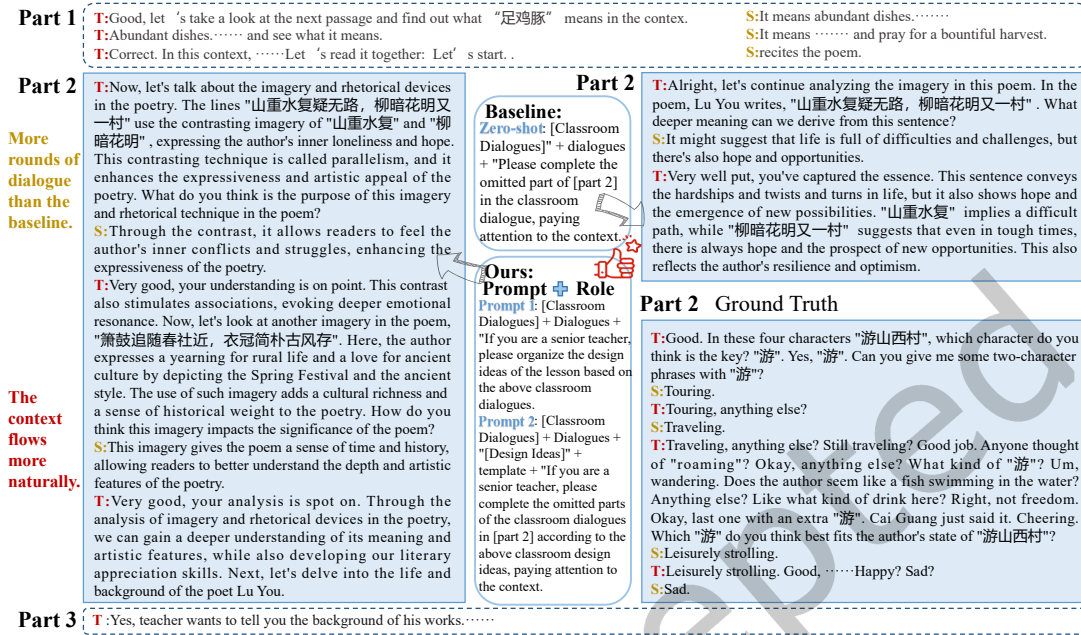


Fig. 3. This example demonstrates a dialogue completion task. Part 1 shows the starting content of the classroom dialogue sample, and part 3 shows the subsequent context of the dialogue. Our task is to complete the missing middle part 2. The baseline method is to use ChatGPT for zero-shot prompting generation, as shown in the blue box on the right. In comparison, our proposed ELF framework performs significantly better, as shown in the blue box on the left. Compared with the baseline method and ground truth, the result of completing the dialogue using the ELF framework has more teacher-student dialogue turns and the content is more coherent, with tighter connections between the upstream and downstream context. Details are discussed in Section 4.1.2.

4.2 Expertise Transfer Learning

4.2.1 Task Description. It is widely known that improving teachers' teaching abilities often requires them to continuously absorb and practice successful teaching methods, thereby honing teaching skills [21, 74]. However, excellence in teaching talent is not distributed evenly, with remote areas particularly lacking. At the same time, masters' classroom designs are usually viewed as benchmarks for teaching quality due to their uniqueness and efficiency. Their distinctive teaching philosophies and styles cannot only spark students' curiosity but also deepen students' understanding and application of knowledge. However, master teachers are inevitably limited in number, meaning most students cannot directly experience the charm of master classrooms. Therefore, how to quickly and accurately learn and inherit the teaching styles of these excellent education practitioners has become an urgent problem in current education.

To address this issue, we conceived the scenario of "Expertise Transfer Learning". The core is to utilize the powerful understanding capabilities of LLM to deeply analyze the essence of existing master teachers' classroom designs and comprehensively and organically extend them to a new topic while maintaining the original teaching wisdom, teaching methods and coherent teaching structure. This educational concept is proposed based on a deep understanding of the thirst for quality educational resources and scientific judgment of the transferability of teaching styles. The excellence of master teachers' classroom designs lies in their profound grasp of knowledge points, precise guidance of student thinking, and skillful regulation of the classroom atmosphere.

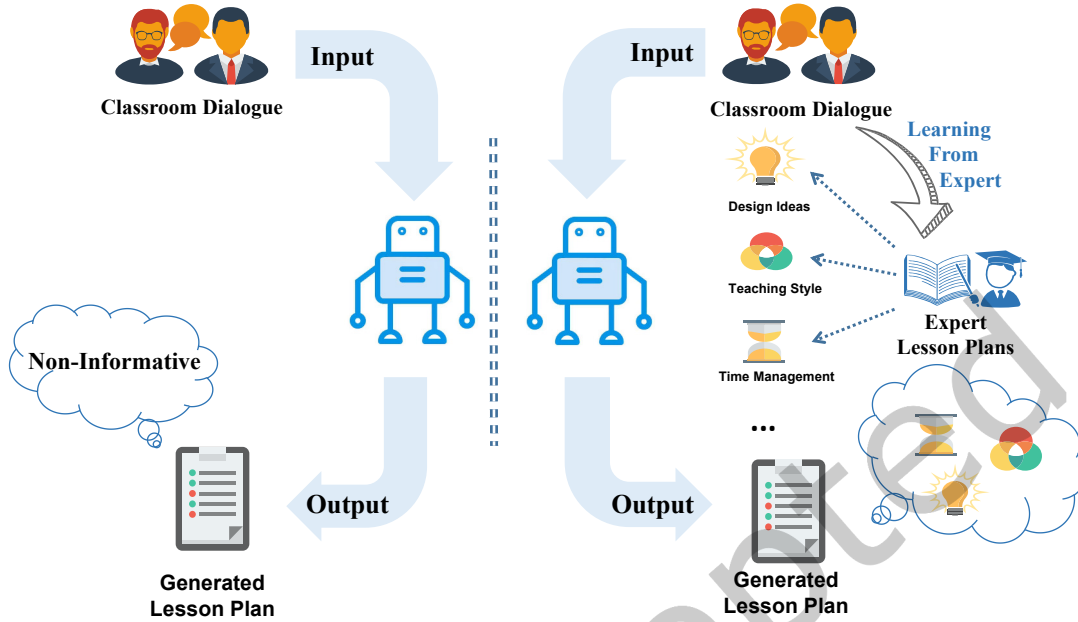


Fig. 4. An overview of the Expertise Transfer Learning task. High-quality instructional design ideas can be generated from automated learning expert templates in our AI-targeted teaching scenarios.

We aim to build an advanced AIGC system that can learn and imitate master teachers' teaching styles to widely disseminate these valuable teaching experiences and benefit more students and teachers. Directly prompting LLM to generate lesson plans is less effective and subject to non-informative outcomes.

Based on the above considerations, we expect the "Expertise Transfer Learning" task to achieve the following specific effects: 1) Effectively learn and accurately imitate master teachers' teaching styles and achieve widespread style transfer; 2) With the assistance of intelligent systems, empower ordinary teachers to design more high-quality classrooms that reach the level of master teachers; 3) Provide students with more diversified, high-quality learning experiences to fully arouse their interest in learning and potential.

We show our proposed pipeline in Fig. 4 of lesson plan generation. We give a concrete example of an expert lesson plan of "Operations of adding and subtracting fractions with different denominators" (Fig. 4 Part-A), which we would like to migrate to the new topic such as "Applications of subtraction" (Fig. 4 Part-B). Guided by an expert lesson plan, the LLM is able to effectively shift the time management, teaching style and design ideas to the newly generated lesson plan (Fig. 4 Part-C).

Besides, the evaluation of the transferred contents should extend beyond mere imitation of master teachers' word choices or phrase structures. More significantly, we seek to evaluate factual accuracy, richness of instructional strategies, and rationality of instructional organization, which will be discussed in detail in Section. 5.

4.2.2 Experimental Setup. In previous research [35], we observed that while the results of zero-shot methods appear acceptable on the surface in the Expertise Transfer Learning task, there are still many issues such as unreasonable time allocation, overly generic teaching content lacking focus, and irregular expressions.

To address these issues, we proposed a new method called "Cross-Context-Guidance (CCG)". The core idea is to first utilize existing master design ideas as templates, and fed as prompt prefixes into LLM. In our experiment, given that there are currently many lightweight and adaptive retrieval methods [47], we can easily provide an expert template for each discipline to better fit the characteristics of each discipline by retrieving from the database. Experimental results showed that the template-based method produced more formally expressed content with a more reasonable time allocation and prominent teaching focus, achieving our expected effects to some degree. To further optimize the quality and professionalism of the

generated content, we also introduced role assignment strategies to allow LLM to generate from the perspective of specific roles for a more educationally valid output.

In addition, we added a disclaimer [33], emphasizing that the provided templates are for reference only and the actual generated content will be adjusted based on the topic. This can make the generated content focus more on the topic expected by the user while learning from the given design ideas in templates to improve the model's flexibility and practicality.

The specific implementation steps of our method are as follows:

Step 1: Topic Identification. First identify the topic for which classroom design ideas need to be generated, which is the foundation of the entire method.

Step 2: Template Matching and Selection. Use existing master design ideas as a template. Then use CCG to feed this template as a prompt prefix into LLM to allow it to learn the master's design ideas.

Step 3: Dynamic Role Assignment. To generate more professional and educationally valid content, we assign LLM a specific role such as "master teacher". When generating content, we will provide prompts such as "Assume you are a master teacher" to guide it to generate from the perspective of a specific role.

Step 4: Task Explanation. Clearly explain to LLM the task it needs to complete, which is to generate design ideas for the new topic. This helps ensure LLM has a clear understanding of the specific task to be completed.

Step 5: Add Disclaimer. After task explanation, we provide a disclaimer emphasizing that the template is for reference only and the actual generated content will be adjusted based on the topic. This aims to prevent LLM from relying too heavily on the template and ignoring the topic we really want to generate.

Step 6: Generation. Based on the previous steps and guidance, we finally have LLM generate design ideas for the new topic.

Figure 5 shows an example of Expertise transfer. By comparing zero-shot questioning and our designed series of methods (CCG + Role Setting + Disclaimer), we can observe that the latter makes it easier for LLM to grasp the logical framework of the given expert lesson plan and achieve outputs of expert quality. Therefore, template transfer (left column) achieved better results. Details of specific comparative results and in-depth analysis will be discussed in Section 5.3.

5 Human Judgement and Automated Scoring of Generated Teaching Content

To better evaluate the text generation quality of our proposed methods, we adopted two methods for evaluation in the tasks of "Dialogue Completion" and "Expertise Transfer Learning". One is to score the coherence of the text through the Large Model (LM), and the other is to distribute questionnaires to obtain human evaluation results. These two methods take into account the objectivity and subjectivity of judging the generated content, which largely avoids significant biases and makes the results more reliable.

5.1 Perplexity-based Statistical Evaluation (PPL)

Text generation plays a crucial role in the field of natural language processing, aiming to generate coherent, accurate, and semantically rich text. In the educational domain, high-quality text generation is essential for effective educational applications. To address this, we propose a metric called Perplexity-based Statistical Evaluation, designed to evaluate the quality of text generated by large-scale educational models. Specifically, we utilize the Perplexity Score (PPL) as its metric, which measures the inverse log likelihood of a sentence in statistical language modeling. In NLP research, PPL is widely adopted to reflect the quality of the generated text [14]. The definition of PPL is described as follows.

For a sentence $s = w_1, w_2, \dots, w_k$, the likelihood of it can be decomposed as the sequential product of the conditional probability of each word as follows:

$$P(s) = P(w_1, w_2, \dots, w_k) = P(w_1) P(w_2 | w_1) \dots P(w_k | w_1, w_2, \dots, w_{k-1}), \quad (1)$$

where $P(w_2 | w_1)$ denotes the probability that the next token is w_2 when the previous token is w_1 . Similarly, $P(w_k | w_1, w_2, \dots, w_{k-1})$ denotes the probability that the next token is w_k when the preceding sentence is $w_1, w_2, \dots, w_{k-1}$.

A classroom design that is more student-centered.

I. Introduction (5 minutes) → A clearer schedule.

1. Use questioning to introduce the lesson topic, have students recall the definition and basic properties of multiplication.
2. Encourage student participation and interaction to create a positive learning environment.

II. Prompt Thinking with a Problem (15 minutes)

1. Pose a prompting question such as "If I have 5 apples and each apple has 3 strawberries, can you express this situation using a multiplication equation?"
2. Guide students to think about whether the order of factors affects the result of multiplication, i.e. the commutativity of multiplication.

III. Exploring Commutativity (25 minutes)

1. Ask students to discuss in small groups and share their ideas with each other.
2. Provide concrete examples and equations such as $5 * 3$ and $3 * 5$, encourage students to discover the commutativity of multiplication through calculating and comparing.
3. Guide students to summarize the pattern: the order of factors does not change the product.

IV. Techniques for Solving Problems (15 minutes)

1. Emphasize learning methods and techniques, such as how to quickly apply the commutativity of multiplication.
2. Teach students how to change the order of factors to obtain equivalent multiplication expressions.

V. Conclusion and Reflection (5 minutes)

1. Praise and encourage students for their active participation and problem solving.
2. Summarize the content of this lesson and reiterate the importance of the commutativity of multiplication.
3. Encourage students to reflect and summarize to reinforce their learning.

Baseline:

Zero-shot: "Please generate classroom design ideas for the classroom topic Multiplication Exchange Law."

Ours:

Prompt + Role

Prompt: Template + "Assuming you are a senior teacher, please generate classroom design ideas for the classroom topic Multiplication Exchange Law based on the template given above. Note that the template given is for reference only and the focus still needs to be on the classroom theme."

1. Introduction of concept: I will explain the meaning of the commutative property of multiplication to students. I might say, "The commutative property of multiplication means that the result of multiplying two numbers is not affected by the order they are multiplied in. No matter what order we multiply them in, the result will be the same."
2. Observation of patterns: I will use some examples to help students observe the pattern of the commutative property of multiplication. For example, I can write down some simple multiplication problems such as $2 * 3$, $4 * 5$, etc. and then have students observe the products and find the rule.
3. Discussion and explanation: I will guide students to discuss the patterns they observed and encourage them to give explanations. Students may say, "Whether it is $2 * 3$ or $3 * 2$, the result is still 6." I will emphasize their observation results and summarize the concept of the commutative property of multiplication together with them.
4. Mathematical proof: To consolidate students' understanding, I will provide a simple mathematical proof to prove the commutative property of multiplication. For example, I can use a physical model to explain it. I can take out 5 of the same apples and divide them into 3 groups with 2 apples in each group. Then I can rearrange the apples so that students can see that no matter if I first divide the 5 apples into 3 groups or first divide them into 2 groups, the final result will be the same.
5. Practice application: To deepen students' understanding of the commutative property of multiplication, I will provide some practice exercises for them to apply the concept. I will encourage students to use the commutative property to verify results in the problem-solving process.
6. Summary and review: At the end of the class, I will summarize the key points of the commutative property of multiplication with students, and encourage them to find more examples in daily life to apply this concept. I may also assign some related homework to help students consolidate their learning.

Fig. 5. This is an example demonstration of Expertise Transfer Learning task. The ChatGPT zero-shot generation (right column) is more inclined to describe the behaviors that teachers need to perform in the classroom when designing lesson plans, which results in a large amount of "I will" expressions in the generated content. The results generated by our ELF framework (left column) provide more directional guidance, and its content and structure are more like a lesson plan carefully designed by educational experts.

The commonly adopted metric perplexity (PPL) in NLP research measures the quality of a generated sentence s as the normalized inverse likelihood of a sequence as follows:

$$PPL(s) = \sqrt[N]{\prod_{i=1}^N \frac{1}{P(w_k | w_1 w_2 \dots w_{k-1})}} = \exp \left\{ -\frac{1}{N} \sum_i \log p_{\theta}(w_k | w_{<k}) \right\}. \quad (2)$$

In practice, we deploy the pre-trained language model GPT-2 [8] and feed the text s to obtain the cross-entropy loss ℓ_s , which coincides with the average negative log-likelihood term of Eq.(2). By taking $PPL(s) = \exp(\ell_s)$, we thus obtain the perplexity value efficiently. In practice, we use the pre-trained GPT-2-Chinese² from Huggingface for language modeling and providing the PPL evaluation metric.

A lower perplexity score means that the language model is more accurate in predicting the next word or phrase in a given text. Since PPL is obtained through a language model, it also reflects that the generated content is consistent with the internal statistical rules of the model, thereby making the generated sentences more coherent, reasonable and natural.

Hence, we employ perplexity (PPL) as a key indicator for assessing the quality of text generation for educational purposes. This statistically-based evaluation method offers significant advantages, reflecting coherence (lower perplexity indicates smoother sentence connections) and considering contextual prediction ability (measuring the overall prediction proficiency within the given textual context).

²<https://huggingface.co/uer/gpt2-chinese-cluecorpussmall>

5.2 Human Evaluation with Questionnaires (HEQ)

Besides using PPL based on statistical language modeling, we also conduct qualitative evaluation through human assessment of the generated content quality.

To ensure comprehensive evaluation, we design a five-point Likert scoring survey. Specifically, we invited 10 human judges and asked them to give a 1-5 score, with higher score indicating better quality, for each question in the questionnaires.

In order to collect high-quality feedback, we selected our human judges from a pool of educators and students specializing in educational fields in our normal school. These participants have undergone formal training in pedagogy and possess a strong understanding of dialogue quality in an educational context. Their expertise allows them to evaluate the AI-generated content effectively, ensuring that the feedback collected is both relevant and informed.

This survey covers two main aspects: 1) Semantic Analysis and 2) Educational Significance, and both of them are split into three specified points. We will explain the question settings in all questionnaires in detail below.

GRUEN [75] is an excellent tool for evaluating generated text, which focuses on content such as grammaticality, non-redundancy, focus, structure, and coherence, suggesting to some extent that language fluency is an important metric for evaluating the quality of text generated. Coherence is also a crucial factor in evaluating text, and research [61] emphasizes that the context must meet coherence criteria to effectively convey the message. Based on the theoretical knowledge mentioned above, we ask the human judges to rate the generated content based on criteria covering 1) Language Smoothness, 2) Clarity of Expression, and 3) Contextual Consistency.

- **Language Smoothness** assesses grammatical correctness and natural expression.
- **Clarity of Expression** gauges the ease with which readers or listeners can understand the content.
- **Contextual Consistency** refers to the coherence and alignment of relevant information or behavior within a specific context.

Additionally, the metric of usefulness mentioned in research [19] subjectively evaluates the practical application of the question in classroom teaching, encompassing its potential use in lectures, assignments, tests, etc. Thus, we designed the educational analysis covering 1) Factual Accuracy, 2) Richness of Instructional Strategies, and 3) Rationality of Instructional Organization to evaluate the usefulness of the generated contents.

- **Factual Accuracy** reflects whether the teaching content strictly adheres to the accuracy and scientific nature of academic knowledge, and whether the knowledge conveyed is consistent with the knowledge system of the discipline.
- **Richness of Instructional Strategies** reflects whether teachers can flexibly apply a variety of teaching methods and means according to students' learning needs and characteristics, such as lecturing, demonstration, discussion, case analysis, role playing and other forms.
- **Rationality of Instructional Organization** reflects the overall planning and design of the teacher's classroom teaching process, which requires teachers to reasonably arrange the sequence and progress of teaching content according to students' cognitive laws and the requirements of teaching objectives, and to ensure that each teaching link is closely connected and orderly.

By providing the five-point scoring based on our designed criteria, we provide a comprehensive assessment of the content generation quality from human judges. The use of human evaluation methods also enhances the objectivity and reliability of the assessment, thereby increasing the credibility of the research.

5.3 Results

In the two key tasks of teaching scenarios, we compare the performance of the baseline method, ground truth and our proposed method in terms of text generation quality.

Baseline. We adopt GPT-3.5-16k³ as the baseline model and follow the zero-shot strategy to prompt for answers.

Our method integrates the two optimization strategies proposed in Section 4 based on baseline method.

Expert Annotation. In the dialogue completion task, we directly used the actual classroom dialogues of the master teacher as the ground-truth expert annotation.

³<https://platform.openai.com/docs/models/gpt-3-5>

Pseudo Annotation. In the expertise transfer learning task, since classroom dialogues can directly reflect the master teacher’s classroom design ideas, we used GPT-4⁴ to generate the classroom design ideas based on the master teacher’s classroom dialogues as *pseudo* annotation.

Table 3. Score results for two tasks (Dialogue Completion, Expertise Transfer Learning) two educational scenarios (Literature, Mathematics) for two metrics (PPL, HEQ). The best performers are **bolded** and the next best are underlined. Through a T-test significance test, results are marked with an asterisk (*) to indicate that the values are significantly better than others at different confidence level. Additionally, Pearson’s correlation coefficient has been utilized to illustrate the relationship between PPL and HEQ.

Method	Metrics	Dialogue Completion			Pearson Correlation r
		Literature	Mathematics	Average	
Baseline	PPL ↓	<u>13.60</u>	<u>10.14</u>	<u>11.87</u>	-0.90
	HEQ ↑	2.38	2.67	2.53	
Expert Annotation	PPL ↓	15.09	10.46	12.78	-0.98
	HEQ ↑	<u>4.23</u>	4.68**	4.46	
ELF (ours)	PPL ↓	12.54	9.34	10.94*	-0.99
	HEQ ↑	4.62**	<u>3.13</u>	<u>3.88</u>	
Method	Metrics	Expert Transfer Learning			Pearson Correlation r
		Literature	Mathematics	Average	
Baseline	PPL ↓	<u>6.78</u>	<u>6.78</u>	<u>6.78</u>	-0.90
	HEQ ↑	2.78	2.55	2.67	
Pseudo Annotation	PPL ↓	7.59	9.22	8.41	-0.93
	HEQ ↑	<u>3.80</u>	<u>4.02</u>	<u>3.91</u>	
ELF (ours)	PPL ↓	6.19*	6.59	6.39	-0.86
	HEQ ↑	4.43*	4.23	4.33**	

Table. 3 presents the detailed comparison results of the baseline method, annotation and ELF (ours) method on the tasks of Dialogue Completion and Expertise Transfer Learning. In Dialogue Completion, our method achieves an average PPL of 10.94 and an average HEQ of 3.88. Further refined analysis, when evaluated on individual datasets, it obtains a PPL of 12.54 and HEQ of 4.62 on the literature dataset, and a PPL of 9.34 and HEQ of 3.13 on the mathematics dataset. In Expertise Transfer Learning, our method achieves an average PPL of 6.39 and an average HEQ of 4.33. When evaluated on individual datasets, it obtains a PPL of 6.17 and HEQ of 4.43 on the literature dataset, and a PPL of 6.59 and HEQ of 4.23 on the mathematics dataset.

These results fully demonstrate the effectiveness and superiority of the proposed ELF method over baselines on different types of datasets for the tasks of Dialogue Completion and Expertise Transfer Learning.

- Our ELF outperforms the baseline method in Dialogue Completion task across both topics of Literature and Mathematics for both PPL and HEQ metrics, achieving a lead of approximately 7.8% in PPL and exceeding by over 1 level in HEQ. Furthermore, the PPL demonstrates a statistically significant improvement over the baseline with a 95% confidence level. Regarding HEQ, while the results are slightly lower than those from expert annotations, this is expected and considered a normal and reasonable performance.

⁴<https://platform.openai.com/docs/models/gpt-4-and-gpt-4-turbo>

- Our ELF also surpasses the baseline method in Expertise Transfer Learning task across both topics, leading about 5.8% in PPL over 1.5 levels higher in HEQ. Notably, the HEQ metric is significantly better than the next best method with a confidence level of 99%.
- The consistency result of our method indicates that our generated texts are both smoother (a smaller PPL) and logistic better (a larger HEQ) than the baseline.
- Especially in Literature, our method significantly outperforms the baseline (ChatGPT) with a HEQ score that was over 2 levels higher (4.62 vs. 2.38) in the Dialogue Completion task, and over 1.5 levels higher (4.43 vs. 2.78) in the Expertise Transfer Learning task. For Math, our method also outperforms the baseline in two tasks, achieving HEQ scores approximately 0.5 and 1.5 levels higher, respectively.
- Our method in HEQ in Dialogue Completion task is slightly lower than Expert Annotation (3.88 vs. 4.46) as expected. However, our method achieves a lower PPL compared with Expert-Annotation (10.94 vs. 12.78), indicating that our generated texts are desired in language modeling, while maybe lacking in richness.
- We verify the Pearson correlation coefficient between PPL and HEQ, which indicates a strong negative correlation between the two metrics ($|r| > 0.8$). This negative correlation is expected as, by definition, a lower PPL signifies better performance while a higher HEQ indicates superior quality. Therefore, the two metrics are in agreement: texts with higher PPL values tend to exhibit lower contextual coherence, resulting in suboptimal performance. This relationship is consistent with our intuitive understanding.
- Another noteworthy finding is that, in the Expertise Transfer Learning task, our ELF demonstrates superior performance in terms of both PPL (6.39 vs. 8.41) and HEQ (4.33 vs. 3.91) compared to Pseudo Annotation. This discrepancy is attributed to the fact that, in the transfer-learning task, Pseudo Annotation is generated through GPT-4 zero-shot prompting, which tends to produce more diverse outputs. In contrast, our cross-contextual learning method transfers expertise from provided templates to new topics, resulting in enhanced dialogic smoothness and understandability. A representative result is illustrated in the attached questionnaire sample.

Given the availability of ground truth for Dialogue Completion task, we expand our experiments to test the performance. For evaluation, we employed common NLP metrics such as BERTScore (F1), ROUGE-L (F1), and BLEU-4. These metrics were selected due to their capabilities in assessing the quality of text generation:

- **BERTScore (F1)**: Utilizes contextual embeddings to evaluate semantic similarity, offering a nuanced and context-aware assessment of dialogue completion quality.
- **ROUGE-L (F1)**: Measures the longest common subsequence, reflecting the overlap between the generated and reference texts, and is particularly effective in capturing the fluency and coherence of the responses.
- **BLEU-4**: Computes 4-gram overlap, which is a widely used metric for evaluating the accuracy of text generation, ensuring that the generated dialogue closely matches the ground truth. These metrics provide a comprehensive evaluation framework for dialogue completion, capturing both semantic and lexical aspects of the task.

Table 4. textbfPerformance Comparison Between Baseline and ELF (Ours) on Dialogue Completion Tasks. The table shows the results for two topics, Literature and Mathematics, using BERTScore (F1), ROUGE-L (F1), and BLEU-4 metrics.

Method	Literature			Mathematics		
	BertScore (F1) ↑	Rouge-L (F1) ↑	BLEU-4 ↑	BertScore (F1) ↑	Rouge-L (F1) ↑	BLEU-4 ↑
Baseline	0.688	0.218	0.099	0.734	0.309	0.219
ELF (Ours)	0.692	0.230	0.107	0.738	0.342	0.223

Table 4 illustrates that our method consistently outperforms the baseline across all metrics for both topics, demonstrating enhanced contextual coherence and text generation quality. In the Literature topic, our method achieved a BERTScore that is 0.6% higher, a ROUGE-L score that is 5.5% higher, and a BLEU-4 score that is 8.1% higher than the baseline. Similarly, in the Mathematics topic, our method achieved a BERTScore that is 0.5% higher, a ROUGE-L score that is 10.7% higher, and a BLEU-4 score that is 1.8% higher. These results highlight the effectiveness of our approach in improving the performance of Dialogue Completion tasks.

5.4 Ablation Studies

We compare the performance of variants of our method in Table 5. The Baseline is the zero-shot prompting with ChatGPT, upon which we add Context Learning (CL) and Dynamic Role Setting (DyRS) techniques.

Table 5. Abation of Contextual Learning and Dynamic Role Setting techniques

Teaching Scenarios	Method	PPL ↓	HEQ ↑
Dialogue Completion	Baseline	11.87	2.53
	+ CL	11.03 (-7.1%)	2.68 (+5.9%)
	+ CL + DyRS (ours)	10.94 (-7.8%)	3.88 (+53.4%)
Expertise Transfer Learning	Baseline	6.78	2.67
	+ CL	6.68 (-1.5%)	4.29 (+60.7%)
	+ CL + DyRS (ours)	6.39 (-5.8%)	4.33 (+62.2%)

As shown in Table 5, in the task of Dialogue Completion, adding CL alone leads to improved performance, with a 7.1% decrease in PPL and a 5.9% increase in HEQ. Furthermore, adding both CL and DyRS even greater gains, significantly decreasing PPL by 7.8% and vastly increasing HEQ by 53.4%.

In the task of expertise transfer learning as well, applying CL alone enhances performance through a 1.5% PPL reduction and a sizable 60.7% HEQ increase. Importantly, combining CL and DyRS delivers substantially better results - decreasing PPL by 5.8% and further boosting HEQ by 62.2%. This clearly demonstrates the effectiveness of leveraging both CL and DyRS together for amplifying conversational effectiveness.

6 Conclusion and Future Challenges

In this paper, we proposed a unified framework that tackles the application of powerful large artificial intelligence models in classical classroom teaching scenarios, making it one of the first efforts to tailor AI-generated content for pedagogical purposes. We focus on two typical use cases of AIGC including 1) dialogue auto-completion and 2) expertise transfer learning.

We evaluate the generated contents thoroughly and comprehensively with two designed evaluation methods, including human judgment with designed questionnaires as well as coherence scoring with LLMs. For the former, we design questionnaires based on semantic relations and relevance to the educational field to obtain flexible evaluation results from humans. For the latter, we encode LLM to obtain coherence and perplexity scores of generated sentences from likelihood logarithmic scores, so that the results can be judged based on the same criteria, which helps objectively evaluate the quality of generated content.

Ultimately, we hope that this paper and our unified Educational LLM Framework (ELF) will have a constructive impact in helping more AI and pedagogical researchers understand and push the frontiers of emerging concepts in LLMs and AIGC. We focus on leveraging a limited amount of expert-annotated data to release the high capacity of the LLMs for educational purposes and applications. Our proposed methodologies include guiding the LLMs with demonstrations, learning contextual representation, dynamic role setting and additional disclaimers. We demonstrate that our methods can provide high-quality classroom teaching dialogues as well as expert instructional design ideas in the disciplines of literature and mathematics. Through extensive evaluations with both classical NLP metric *PPL* and human evaluation in terms of language smoothness, clarity of expression, and contextual consistency, we confirm the ability of *ELF* to improve the quality of educational data generation, which in turn provides strong guidance for more effective application of LLMs and AIGC.

The future challenge lies in how to generate rich multimedia content such as figures, videos, and soundtracks. This capability is crucial to improving interactivity and engagement in educational materials. By advancing the ability of LLMs to manage and integrate various media types, we can offer more dynamic and comprehensive learning experiences. Recent studies [17, 62] proposing methods to handle data from multi-annotated datasets using collaborative learning and local

semantic analysis could also inspire educational data analysis in various scenarios. Another significant challenge is how to incorporate personalization to enhance AIGC. By analyzing and assessing the learning situation of each student, AI can tailor educational content to meet individual characteristics and requirements. This customization ensures a more effective and enriched learning experience, catering to the unique needs of every student.

References

- [1] accessed May 2, 2023. Midjourney. <https://www.midjourney.com/>.
- [2] R. Abdelghani, Y. H. Wang, X. Yuan, et al. 2024. GPT-3-Driven Pedagogical Agents to Train Children’s Curious Question-Asking Skills. *International Journal of Artificial Intelligence in Education* 34, 1 (2024), 483–518. doi:10.1007/s40593-023-00340-7
- [3] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altmenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- [4] Robin J Alexander. 2001. Border crossings: Towards a comparative pedagogy. *Comparative Education* 37, 4 (2001), 507–523.
- [5] Christa Asterhan, Sherice Clarke, and Lauren Resnick. 2015. Socializing intelligence through academic talk and dialogue. *Socializing Intelligence Through Academic Talk and Dialogue* (2015), 1–480.
- [6] Nathaniel Blanchard, Michael Brady, Andrew M Olney, Marci Glaus, Xiaoyi Sun, Martin Nystrand, Borhan Samei, Sean Kelly, and Sidney D’Mello. 2015. A study of automatic speech recognition in noisy classroom environments for automated dialog analysis. In *Artificial Intelligence in Education: 17th International Conference, AIED 2015, Madrid, Spain, June 22-26, 2015. Proceedings* 17. Springer, 23–33.
- [7] Adam Bouyamourn. 2023. Why LLMs Hallucinate, and How to Get (Evidential) Closure: Perceptual, Intensional, and Extensional Learning for Faithful Natural Language Generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 3181–3193. doi:10.18653/v1/2023.emnlp-main.192
- [8] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [9] Yihan Cao, Siyu Li, Yixin Liu, Zhiling Yan, Yutong Dai, Philip S Yu, and Lichao Sun. 2023. A comprehensive survey of ai-generated content (aigc): A history of generative ai from gan to chatgpt. *arXiv preprint arXiv:2303.04226* (2023).
- [10] Bo Cheng, Jia Zhu, and Xiaoyong Hu. 2022. Exploring the collective process of classroom dialogue using sequential pattern mining technique. *International Journal of Educational Research* 115 (2022), 102050.
- [11] Davide Colla, Matteo Delsanto, Marco Agosto, Benedetto Vitiello, and Daniele P Radicioni. 2022. Semantic coherence markers: The contribution of perplexity metrics. *Artificial Intelligence in Medicine* 134 (2022), 102393.
- [12] Fuwei Cui, Qian Cui, and Yongduan Song. 2020. A survey on learning-based approaches for modeling and classification of human-machine dialog systems. *IEEE Transactions on Neural Networks and Learning Systems* 32, 4 (2020), 1418–1432.
- [13] Ruiguo Cui and Peter Teo. 2021. Dialogic education for classroom teaching: a critical review. *Language and education* 35, 3 (2021), 187–203.
- [14] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Tamar Solorio (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 4171–4186. doi:10.18653/v1/N19-1423
- [15] Ramon Dijkstra, Zülküf Genç, Subhadeep Kayal, Jaap Kamps, et al. 2022. Reading Comprehension Quiz Generation using Generative Pre-trained Transformers.. In *iTextbooks@ AIED*. 4–17.
- [16] Songtao Ding, Shiru Qu, Yuling Xi, and Shaohua Wan. 2020. Stimulus-driven and concept-driven analysis for image caption generation. *Neurocomputing* 398 (2020), 520–530.
- [17] Songtao Ding, Hongyu Wang, Hu Lu, Michele Nappi, and Shaohua Wan. 2022. Two path gland segmentation algorithm of colon pathological image based on local semantic guidance. *IEEE Journal of Biomedical and Health Informatics* 27, 4 (2022), 1701–1708.
- [18] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130* (2024).
- [19] Sabina Elkins, Ekaterina Kochmar, Iulian Serban, and Jackie CK Cheung. 2023. How useful are educational questions generated by large language models?. In *International Conference on Artificial Intelligence in Education*. Springer, 536–542.
- [20] Aditya Ramesh et al. 2022. Hierarchical Text-Conditional Image Generation with CLIP Latents.
- [21] Shaoxiong Feng, Xuancheng Ren, Hongshen Chen, Bin Sun, Kan Li, and Xu Sun. 2020. Regularizing Dialogue Generation by Imitating Implicit Scenarios. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 6592–6604. doi:10.18653/v1/2020.emnlp-main.534
- [22] Tao Feng, Zifeng Wang, and Jimeng Sun. 2023. Citing: Large language models create curriculum for instruction tuning. *arXiv preprint arXiv:2310.02527* (2023).

- [23] Leon A Gatys, Alexander S Ecker, and Matthias Bethge. 2016. Image style transfer using convolutional neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2414–2423.
- [24] Irina Gruzdo, Iryna Kyrychenko, Glib Tereshchenko, and Olga Cherednichenko. 2020. Application of Paragraphs Vectors Model for Semantic Text Analysis.. In *COLINS*. 283–293.
- [25] Valentina Guzmán and Antonia Larrain. 2021. The transformation of pedagogical practices into dialogic teaching: towards a dialogic notion of teacher learning. *Professional Development in Education* (2021), 1–14.
- [26] Namgyu Ho, Laura Schmid, and Se-Young Yun. 2023. Large Language Models Are Reasoning Teachers. arXiv:2212.10071 [cs.CL]
- [27] Christine Howe and Manzoorul Abedin. 2013. Classroom dialogue: A systematic review across four decades of research. *Cambridge journal of education* 43, 3 (2013), 325–356.
- [28] Christine Howe, Sara Hennessy, Neil Mercer, Maria Vrikki, and Lisa Wheatley. 2019. Teacher–Student Dialogue During Classroom Teaching: Does It Really Impact on Student Outcomes? *Journal of the Learning Sciences* 0 (03 2019), 1–51.
- [29] Christine Howe, Sonia Ilie, Paula Guardia, Riikka Hofmann, Neil Mercer, and Fran Riga. 2015. Principled improvement in science: Forces and proportional relations in early secondary-school teaching. *International Journal of Science Education* 37, 1 (2015), 162–184.
- [30] Christine Howe and Neil Mercer. 2017. Commentary on the papers. *Language and Education* 31, 1 (2017), 83–92.
- [31] Enkelejda Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, Daryna Dementieva, Frank Fischer, Urs Gasser, Georg Groh, Stephan Günnemann, Eyke Hüllermeier, et al. 2023. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and individual differences* 103 (2023), 102274.
- [32] David Liljequist, Britt Elfving, and Kirsti Skavberg Roaldsen. 2019. Intraclass correlation—A discussion and demonstration of basic features. *PloS one* 14, 7 (2019), e0219854.
- [33] Xiangyang Liu, Tianqi Pang, and Chenyou Fan. 2023. Federated Prompting and Chain-of-Thought Reasoning for Improving LLMs Answering. In *Knowledge Science, Engineering and Management*, Zhi Jin, Yuncheng Jiang, Robert Andrei Buchmann, Yaxin Bi, Ana-Maria Ghiran, and Wenjun Ma (Eds.). Springer Nature Switzerland, Cham, 3–11.
- [34] Zeyi Liu, Arpit Bahety, and Shuran Song. 2023. REFLECT: Summarizing Robot Experiences for Failure Explanation and Correction. In *7th Annual Conference on Robot Learning*. https://openreview.net/forum?id=8yTS_nAILxt
- [35] Chung Kwan Lo, Khe Foon Hew, and Morris Siu-yung Jong. 2024. The influence of ChatGPT on student engagement: A systematic review and future research agenda. *Computers & Education* (2024), 105100.
- [36] Shawn Loewen and Masatoshi Sato. 2018. Interaction and instructed second language acquisition. *Language teaching* 51, 3 (2018), 285–329.
- [37] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2024. SELF-REFINE: iterative refinement with self-feedback. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (*NIPS '23*). Curran Associates Inc., Red Hook, NY, USA, Article 2019, 61 pages.
- [38] Neil Mercer. 2008. The seeds of time: Why classroom dialogue needs a temporal analysis. *The journal of the learning sciences* 17, 1 (2008), 33–59.
- [39] Neil Mercer. 2010. The analysis of classroom talk: Methods and methodologies. *British journal of educational psychology* 80, 1 (2010), 1–14.
- [40] Neil Mercer and Lyn Dawes. 2014. The study of talk between teachers and students, from the 1970s until the 2010s. *Oxford review of education* 40, 4 (2014), 430–445.
- [41] Neil Mercer, Sara Hennessy, and Paul Warwick. 2019. Dialogue, thinking together and digital technology in the classroom: Some educational implications of a continuing line of inquiry. *International Journal of Educational Research* 97 (2019), 187–199.
- [42] George A Miller. 1995. WordNet: a lexical database for English. *Commun. ACM* 38, 11 (1995), 39–41.
- [43] Steven Moore, Huy A Nguyen, Norman Bier, Tanvi Domadia, and John Stamper. 2022. Assessing the quality of student-generated short answer questions using GPT-3. In *European conference on technology enhanced learning*. Springer, 243–257.
- [44] Nikahat Mulla and Prachi Gharpure. 2023. Automatic question generation: a review of methodologies, datasets, evaluation metrics, and applications. *Progress in Artificial Intelligence* 12, 1 (2023), 1–32.
- [45] OpenAI and Josh Achiam .et. 2024. GPT-4 Technical Report. arXiv:2303.08774 [cs.CL] <https://arxiv.org/abs/2303.08774>
- [46] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2024. Training language models to follow instructions with human feedback. In *Proceedings of the 36th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (*NIPS '22*). Curran Associates Inc., Red Hook, NY, USA, Article 2011, 15 pages.
- [47] Tianqi Pang, Kehui Tan, Yujun Yao, Xiangyang Liu, Fanlong Meng, Chenyou Fan, and Xiaofan Zhang. 2024. REMED: Retrieval-Augmented Medical Document Query Responding with Embedding Fine-Tuning. *IJCNN*.
- [48] Ilda Nadia Monica de la Asuncion Pari-Bedoya, Alfonso Renato Vargas-Murillo, and Francisco de Jesús Guevara-Soto. 2023. Challenges and opportunities of AI-assisted learning: a systematic literature review on the impact of ChatGPT usage in higher education. (2023).

- [49] Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. 2023. DreamFusion: Text-to-3D using 2D Diffusion. In *Proceedings of the International Conference on Learning Representations*. <https://openreview.net/forum?id=FjNys5c7VyY>
- [50] Götz Schwab. 2011. From dialogue to multilogue: A different view on participation in the English foreign-language classroom. *Classroom Discourse* 2, 1 (2011), 3–19.
- [51] Jianhao Shen, Yichun Yin, Lin Li, Lifeng Shang, Xin Jiang, Ming Zhang, and Qun Liu. 2021. Generate & Rank: A Multi-task Framework for Math Word Problems. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (Eds.). Association for Computational Linguistics, Punta Cana, Dominican Republic, 2269–2279. doi:10.18653/v1/2021.findings-emnlp.195
- [52] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2024. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems* 36 (2024).
- [53] Patrick E Shrout and Joseph L Fleiss. 1979. Intraclass correlations: uses in assessing rater reliability. *Psychological bulletin* 86, 2 (1979), 420.
- [54] John Sinclair and Malcolm Coulthard. 2013. Towards an analysis of discourse. In *Advances in spoken discourse analysis*. Routledge, 7–40.
- [55] Yu Song, Tianyong Hao, Zhinan Liu, and Zixin Lan. 2020. A systematic review of frameworks for coding towards classroom dialogue. In *Emerging Technologies for Education: 4th International Symposium, SETE 2019, Held in Conjunction with ICWL 2019, Magdeburg, Germany, September 23–25, 2019, Revised Selected Papers* 4. Springer, 226–236.
- [56] Yu Song, Shunwei Lei, Tianyong Hao, Zixin Lan, and Ying Ding. 2021. Automatic classification of semantic content of classroom dialogue. *Journal of Educational Computing Research* 59, 3 (2021), 496–521.
- [57] Yu Song and Jan D Vermunt. 2021. A comparative study of learning patterns of secondary school, high school and college students. *Studies in Educational Evaluation* 68 (2021), 100958.
- [58] Yu Song, Shu Zhang, and Bingman Liu. 2023. Investigating the dialogic patterns of mathematics lessons in different stages of education. *The Journal of Educational Research* 116, 2 (2023), 77–89.
- [59] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971* (2023).
- [60] Immanuel Trummer. 2024. Large Language Models: Principles and Practice. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)*. IEEE, 5354–5357.
- [61] Hakan Ülper. 2011. An evaluation of student texts in the context of Coherence. *Journal of Turkish Studies* 6 (2011).
- [62] Hongyu Wang, Dandan Zhang, Jun Feng, Lucia Cascone, Michele Nappi, and Shaohua Wan. 2024. A multi-objective segmentation method for chest X-rays based on collaborative learning from multiple partially annotated datasets. *Information Fusion* 102 (2024), 102016.
- [63] Shen Wang, Tianlong Xu, Hang Li, Chaoli Zhang, Joleen Liang, Jiliang Tang, Philip S Yu, and Qingsong Wen. 2024. Large language models for education: A survey and outlook. *arXiv preprint arXiv:2403.18105* (2024).
- [64] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. arXiv:2203.11171 [cs.CL]
- [65] Zichao Wang, Andrew Lan, and Richard Baraniuk. 2021. Math Word Problem Generation with Mathematical Consistency and Problem Context Constraints. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (Eds.). Association for Computational Linguistics, Online and Punta Cana, Dominican Republic, 5986–5999. doi:10.18653/v1/2021.emnlp-main.484
- [66] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2024. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems (New Orleans, LA, USA) (NIPS '22)*. Curran Associates Inc., Red Hook, NY, USA, Article 1800, 14 pages.
- [67] Qifeng Wei and Wangyue Qi. [n. d.]. Research on Innovative Teaching Models in Accounting Education Based on Artificial Intelligence Generated Content (AIGC). ([n. d.]).
- [68] Jingfeng Yang, Hongye Jin, Ruixiang Tang, Xiaotian Han, Qizhang Feng, Haoming Jiang, Shaochen Zhong, Bing Yin, and Xia Hu. 2024. Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond. *ACM Trans. Knowl. Discov. Data* 18, 6, Article 160 (apr 2024), 32 pages. doi:10.1145/3649506
- [69] Shi Yang, Siyuan Yang, Chaoran Tong, et al. 2023. In-Depth Application of Artificial Intelligence-Generated Content AIGC Large Model in Higher Education. *Adult and Higher Education* 5, 19 (2023), 9–16.
- [70] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik R Narasimhan. 2023. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. In *Thirty-seventh Conference on Neural Information Processing Systems*. <https://openreview.net/forum?id=5Xc1ecxO1h>
- [71] Weijiang Yu, Yingpeng Wen, Fudan Zheng, and Nong Xiao. 2021. Improving math word problems with pre-trained knowledge and hierarchical reasoning. In *Proceedings of the 2021 conference on empirical methods in natural language processing*. 3384–3394.

- [72] Peng Zhang and Gemma Tur. 2024. A systematic review of ChatGPT use in K-12 education. *European Journal of Education* 59, 2 (2024), e12599.
- [73] Wayne Xin Zhao, Kun Zhou, et al. 2023. A Survey of Large Language Models. *arXiv preprint arXiv:2303.18223* (2023).
- [74] Jiangyuan Zhou and Wei Guo. 2016. Imitation in Undergraduate Teaching and Learning. *The Journal of Effective Teaching* 16 (2016), 5–27.
- [75] Wanzheng Zhu and Suma Bhat. 2020. GRUEN for Evaluating Linguistic Quality of Generated Text. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: Findings*. 94–108.

Received 28 January 2024; revised 10 November 2024; accepted 22 December 2024